

	Attention																								
	Hidden input	Q projection		Q RoPE						K projection		K RoPE					K repeat	K layout		V projection		V head layout	V repeat	V matmul layout	Q x K^T
	layout	layout	matmul	transpose	mul	negf	mul	add	layout	layout	matmul	transpose	mul	negf	mul	add	expand	transpose	layout	layout	matmul	transpose	expand	layout	matmul
edge / prefill / B=1 / L=128	C0	C0	C0	C1	C4	C5	C6	C6	C11	C2	C2	C2	C7	C8	C9	C9	C9	C11	C11	C3	C3	C3	C10	C14	C11
edge / prefill / B=1 / L=4096	C0	C1	C2	C3	C10	C11	C12	C12	C18	C4	C5	C6	C13	C14	C15	C15	C15	C17	C19	C7	C8	C9	C16	C22	C20
server / prefill / B=8 / L=128			C0	C0	C3	C4	C5	C5			C1	C1	C6	C7	C8	C8	C8	C10			C2	C2	C9		C10
server / prefill / B=8 / L=4096			C0	C0	C3	C4	C5	C5			C1	C1	C6	C7	C8	C8	C8	C10			C2	C2	C9		C11
edge / decode / B=1 / L=128	C0	C1	C1	C1	C3	C4	C5	C5	C10	C2	C2	C2	C6	C7	C8	C8	C8	C10	C10	C1	C1	C1	C9	C13	C10
edge / decode / B=1 / L=4096	C0	C1	C1	C1	C3	C4	C5	C5	C11	C2	C2	C2	C6	C7	C8	C8	C9	C11	C11	C1	C1	C1	C10	C14	C11
server / decode / B=8 / L=128	C0		C0	C0	C2	C3	C4	C4	C9		C1	C1	C5	C6	C7	C7	C7	C9			C1	C1	C8		C9
server / decode / B=8 / L=4096	C0		C0	C0	C2	C3	C4	C4	C9		C1	C1	C5	C6	C7	C7	C7	C9			C1	C1	C8		C9

	Attention (continued)															FFN											
	Scale	Add mask	FP32 extend	Max reduce	Subtract max	Exp	Sum reduce	Divide	BF16 trunc	Output layout	Attention x V	O projection				Hidden input	Gate projection		SiLU		FFN Up projection		Elementwise x	Down projection			
	trunc mul	add	extend	max reduce	sub	exp	sum reduce	div	trunc	layout	matmul	transpose	layout	layout	matmul	layout	layout	matmul	sigmoid	mul	layout	matmul	mul	layout	layout	matmul	
edge / prefill / B=1 / L=128	C11	C11	C12	C12	C13	C13	C13	C13	C13	C13	C14	C15	C15	C15	C15	C0	C1	C1	C2	C2	C3	C3	C4	C4	C5	C5	
edge / prefill / B=1 / L=4096	C20	C20	C20	C20	C21	C21	C21	C21	C21	C21	C22	C23	C23	C24	C25	C0	C1	C2	C3	C3	C4	C5	C6	C6	C7	C8	
server / prefill / B=8 / L=128	C10	C10	C10	C10	C11	C11	C11	C11	C11		C11	C12			C12			C0	C0	C0		C0	C0			C1	
server / prefill / B=8 / L=4096	C11	C11	C11	C11	C12	C12	C12	C12	C12		C13	C13			C14			C0	C0	C0		C0	C0			C1	
edge / decode / B=1 / L=128	C11	C11	C11	C11	C12	C12	C12	C12	C12	C12	C13	C13	C14	C14	C14	C0	C1	C1	C1	C1	C1	C1	C2	C2	C3	C3	
edge / decode / B=1 / L=4096	C12	C12	C12	C12	C13	C13	C13	C13	C13	C13	C14	C14	C15	C15	C15	C0	C1	C1	C1	C1	C1	C1	C2	C2	C3	C3	
server / decode / B=8 / L=128	C10	C10	C10	C10	C11	C11	C11	C11	C11	C12	C12	C12	C13		C13	C0		C1	C1	C1		C1	C1	C1		C2	
server / decode / B=8 / L=4096	C10	C10	C10	C10	C11	C11	C11	C11	C11	C12	C12	C12	C13		C13	C0		C1	C1	C1		C1	C1	C1		C2	